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A deep learning framework for finger motion recognition using forearm ultrasound imaging

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Precise classification of finger movements is essential for effective hand gesture recognition, as it facilitates reliable control across multiple degrees of freedom. Surface electromyography (sEMG) has been widely used for monitoring muscle activities in the forearm and applied for the classification of finger motion. However, due to its inherent limitations, sEMG fails to provide a satisfactory solution for finger motion recognition. As an alternative, A-mode ultrasound-based sensing methods have been proposed and have demonstrated great potential. In this paper, we present a finger motion classification method based on deep learning using forearm ultrasound imaging. Since B-mode ultrasound offers 2D visualization of muscle activity, it capture broader anatomical features and reducing sensitivity to transducer placement compared to A-mode ultrasound. Real-time ultrasound images of forearm muscles were acquired during nine predefined finger motions, comprising five single-finger motions and four multi-finger motions. A framework was developed to classify the finger motion. The framework demonstrated high accuracy in recognizing finger motions, with an overall average classification accuracy and an F1 score of 95.64% and 0.9563, respectively. The experimental results demonstrated the feasibility of the proposed method in accurately classifying various finger movements. The proposed method holds potential for various applications such as gesture recognition in virtual reality (VR)/augmented reality (AR), controlling robotic hands, and assisting in medical rehabilitation for finger dexterity.

Keywords Deep learning, Finger motion recognition, Human-machine interface, Prostheses control, Ultrasound imaging

Abbreviations

BR	Brachioradialis
APL	Abductor pollicis longus
EPL	Extensor pollicis longus
FCU	Flexor carpi ulnaris
FCR	Flexor carpi radialis
FDP	Flexor digitorum profundus
FDS	Flexor digitorum superficialis
FPL	Flexor pollicis longus
PL	Palmaris longus
PT	Pronator teres
RA	Radial artery
MN	Median nerve
UA	Ulnar artery
UN	Unar nerve

Human-machine interfaces (HMIs) based on non-invasive methods have drawn much attention due to its diverse applications, including rehabilitation engineering, prosthesis control, remote manipulation, and virtual reality

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(VR)/augmented reality (AR)[1–7]. To achieve accurate and reliable hand gesture recognition, various technologies have been employed. Among them, surface electromyography (sEMG), which measures the electrical activity associated with motor unit action potentials of a muscle from the body surface, stands as the most commonly used method for sensing muscle activity in the forearm. Nevertheless, sEMG possesses inherent limitations, including a low signal-to-noise ratio and a lack of specificity for deep muscles[8,9]. On the other hand, ultrasound imaging (or sonography) can visualize the anatomy of muscles and detect changes in both superficial and deep muscles with high spatial resolution in real-time. Thus, ultrasound imaging could be a more suitable alternative to sEMG for sensing muscle activity.

A pioneering study investigated the use of ultrasound imaging on the forearm to identify morphological changes in muscles for powered prosthesis development, revealing a linear relationship between wrist extension angle and extensor muscle morphology[10]. In this work, ultrasound images of the forearm muscles were collected from participants, demonstrating its potential for musculoskeletal control. Due to its simplicity and low cost, one-dimensional (1D) A-mode ultrasound was first investigated as an alternative to sEMG to assess muscle activity. Guo et al. utilized A-mode ultrasound for tracking muscle movements and demonstrated that this method can reliably track wrist movements at various speeds[11]. In addition, three machine learning methods were used to predict wrist angles using 1D A-mode ultrasound, showing potentials in controlling one-degree-of-freedom prosthesis[12]. More recently, four ultrasonic transducers were used to acquire A-mode ultrasound from forearm muscles to predict dexterous musculoskeletal movements[13]. While A-mode ultrasound showed potential for prosthesis control, the accuracy of prediction depends on the positioning of transducers for assessing forearm muscles associated with finger motion. Moreover, the relocation of forearm muscles during pronation and supination (see Fig. 1) poses a significant challenge in the utilization of A-mode ultrasound for prosthesis control.

For hand gesture recognition, accurate classification of finger motions serves as a fundamental prerequisite to achieve precise finger motion control due to its ability to provide multi-degree-of-freedom control. Since B-mode or two-dimensional (2D) images offer more comprehensive information on the forearm muscle activity than other methods, it can effectively visualize the specific muscles in the forearm. Castellini et al. showed a linear relationship between forearm muscle activity and finger position[14]. Subsequently, ultrasound imaging acquired using a mechanically scanned single-element transducer was utilized to evaluate forearm muscle activity[15,16]. In this work, low-resolution binary activity patterns, which were generated through pixel-wise comparison between two consecutive images, were employed as a robust and practical solution in ultrasound-

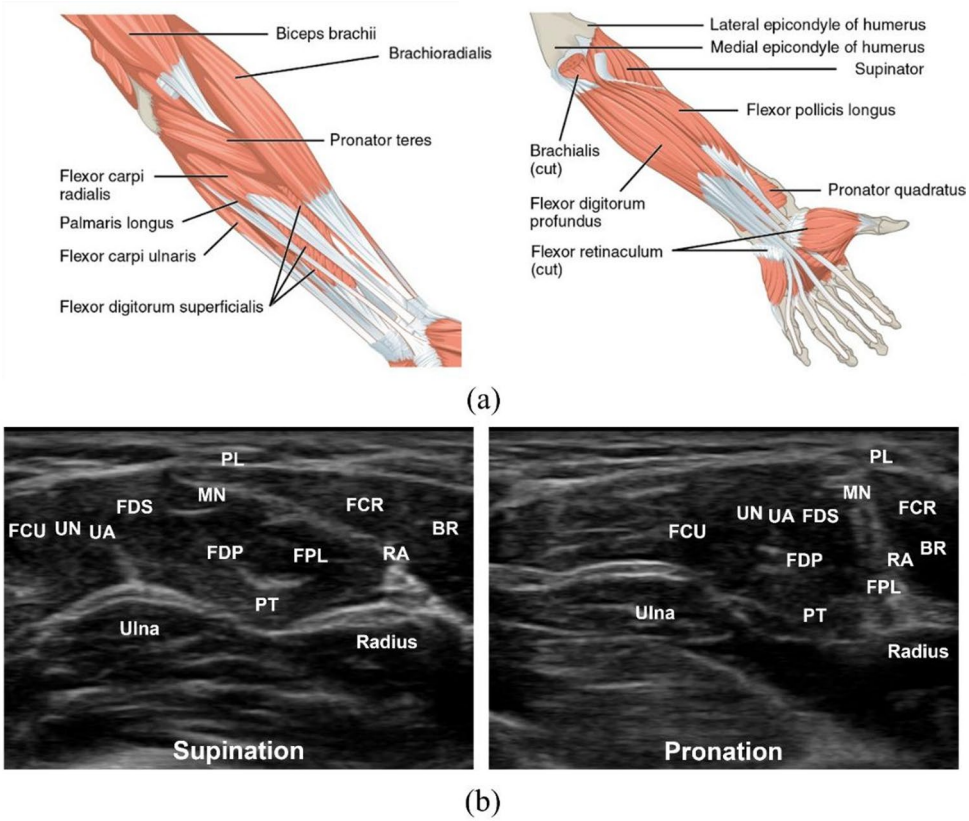


Fig. 1. (**a**) The distribution and structure of the forearm muscles (palmar view) and (**b**) ultrasound image of forearm muscles in pronation and supination positions. The main muscles related to finger motions are annotated. Shifting in muscle positioning by wrist rotation (pronation and supination) is identified in ultrasound images.

based HMI. Huang et al. compared the performance of finger motion classification using both sEMG and ultrasound, demonstrating that A-mode ultrasound achieved a higher average accuracy (95.88%) compared to sEMG (90.14%)[17]. In this work, only 5 columns of ultrasound image were manually selected, i.e., A-mode signal from B-mode image. Principal component analysis (PCA) was applied to reduce the dimension of the original feature space. Subsequently, linear discriminant analysis (LDA) was employed for classifying the finger motion. From the same research group, a prediction of wrist and hand motion using ultrasound imaging was proposed[18]. They could achieve the highest average accuracies of 79.7% for finger motion classification using muti-task convolutional neural network (MTCNN) and 92.5% for wrist rotation classification using subclass discriminant analysis (SDA), respectively.

Unlike A-mode ultrasound, which relies on one-dimensional signals acquired along a fixed position, B-mode ultrasound provides a comprehensive two-dimensional visualization of muscle structures and their dynamic activity. This expanded spatial context enables the system to capture broader anatomical and functional features, reducing sensitivity to variations in transducer placement. In A-mode systems, even slight deviations in probe position or orientation can significantly affect signal characteristics, leading to inconsistent performance[13,15]. In contrast, B-mode imaging captures a wider field of view and retains spatial coherence across frames, thereby maintaining classification accuracy despite variations in wrist posture. This inherent robustness to transducer positioning introduces a key advantage in real applications, where precise and repeatable placement may be difficult to achieve. Consequently, the use of B-mode ultrasound represents a novel and significant step toward more reliable muscle activity analysis. While traditional machine learning methods relying on handcrafted features are limited in modeling the complex spatiotemporal patterns of ultrasound signals, deep learning models can automatically extract hierarchical representations and temporal dynamics, leading to improved classification accuracy and robustness. Furthermore, deep learning enhances generalization across subjects and resilience to variations in probe placement, both of which are essential for practical gesture recognition. Recent advances in miniaturized ultrasound systems further support the feasibility of wearable ultrasound-based hand gesture recognition devices.

Recently, B-mode ultrasound imaging has been investigated for hand gesture classification[19–21]. In those studies, two-dimensional ultrasound images were analyzed using convolutional neural networks (CNNs), which were employed either to extract relevant features subsequently used for machine learning-based classification or to directly classify finger or hand motions at specific time points. These approaches demonstrated the potential of CNNs for capturing spatial features from ultrasound images, but were limited in their ability to fully exploit the temporal dynamics inherent in muscle activity.

This paper presents deep learning-based finger motion classification using B-mode ultrasound imaging. In the work, we introduced a framework based on model ensembling to recognize finger motion utilizing deep learning models developed for action recognition. The proposed framework successfully showed its capability to predict nine dexterous finger motions in both pronation and supination positions of the arm. The results indicate that the proposed method holds promise for achieving precise finger motion control in prostheses and remote control systems.

Methods

Forearm muscles involve in finger movements

Figure 1 depicts the distribution and structure of the forearm muscles (palmar view) and ultrasound image of forearm muscles in pronation and supination positions. These muscles play a crucial role in controlling individual digits[22]. Specifically, FDP, FDS, and EDC enable flexion in the proximal and distal interphalangeal joints of the index, middle, ring, and little fingers, respectively. However, mechanical and neural interconnections in both muscles can cause unintended flexion of neighboring digits when attempting to flex a specific digit, i.e., middle, ring, and little fingers. Control of the thumb involves distinct muscles: FPL, EPL, and APL. Since certain muscles related to finger motion lie in deep tissue, sEMG often fails in accurate measurement of muscle activity. However, as illustrated in Fig. 1, ultrasound imaging readily visualized the flexors and their compartments, enabling real-time analysis of active contraction patterns and passive relaxation according to finger motion. In addition, wrist rotations cause shifts in muscle positioning, indicating the importance of acquiring ultrasound images for both pronation and supination positions to establish a feasible solution for deep-learning-based finger motion recognition.

A deep learning framework for finger motion recognition

We classified nine dexterous finger motions from ultrasound video clips. For stable prediction, we employed model ensembling, in which the results of multiple classifiers are combined. The multiple classifiers included ResNet 3D, ResNet MC, ResNet (2 + 1)D, MViT, and Swin3D, which were specifically designed to handle video clips. In the proposed method, each classifier makes a decision and votes, yielding the final decision. The classifiers are detailed as follows:

1) ResNet 3D: This classifier effectively extracts local and short-term spatiotemporal features using 3D convolution layers[23]. In each layer, spatiotemporally invariant filters are convolved in both spatial and temporal dimensions. As a result, this classifier effectively captures the information encoded in multiple adjacent frames, but it suffers from high computational complexity.
2) ResNet MC: This is devised to reduce the complexity of ResNet 3D, based on the hypothesis that 3D convolution layers are particularly useful in the early layers. Thus, its architecture starts with 3D convolutions to extract spatiotemporal features and switches to using 2D convolutions. In several video recognition tasks, it has outperformed ResNet 3D[24].

3) ResNet (2 + 1)D: This classifier employed spatiotemporal decomposition, which factorizes each 3D convolution layer into spatial (2D) and temporal (1D) convolution layers[24]. With these separate layers, it performs the prediction not only effectively but also efficiently compared with ResNet 3D.
4) MViT: Recently, the vision transformer (ViT) has shown promising performance in computer vision tasks with a global attention mechanism[25]. MViT, based on ViT, yields a multiscale pyramid of features with the QKV pooling operation. Various experiments have demonstrated its effectiveness for video understanding[26].
5) Swin3D: This classifier uses the 3D Swin transformer blocks to take advantage of the inherent spatiotemporal locality of video clips[27]. Note that these blocks are used to introduce the inductive biases of locality, hierarchy, and translation invariance. This classifier leads to a better speed-accuracy trade-off compared to the conventional ViT-based methods.

Figure 2 shows the architectures of five classifiers employed to analyze forearm muscle activity for predicting finger motion. Here, ResNet 3D, MC, and (2 + 1)D are based on a ResNet-18[28], which ensures a fair comparison in terms of model capacity. MViT and Swin3D are designed to have parameters similar to those of ResNet 3D. Despite having similar architectures and comparable model capacities, these models employ distinct mechanisms for temporal information processing. We trained each classifier from scratch by minimizing the cross-entropy loss function. We trained each classifier from scratch by minimizing the cross-entropy loss function with four NVIDIA RTX 4090 GPUs. For optimization, we utilized the AdamW optimizer[29] with an initial learning rate of 0.0001. The learning rate was reduced by a factor of 10 using a scheduler once learning stagnated. The epoch and batch size were 80 and 32, respectively. On average, the training took approximately 3.53 h for ResNet 3D, 3.81 h for ResNet MC, 7.15 h for ResNet (2 + 1)D, 5.63 h for MViT, and 4.58 h for Swin 3D. After training each classifier, we assigned weights $\{w_i\}_{i=1}^{5}$ to five classifiers based on their validation accuracies. Specifically, the validation accuracies $\{v_i\}_{i=1}^{5}$ are normalized by exponential function as

$$w_i = \frac{e^{v_i}}{\sum_{i=1}^{5} e^{v_i}} \tag{1}$$

Therefore, classifiers with higher accuracies have larger weight values. With these weights, we aggregate the prediction results, making the final decision. Without weighted voting, the predictions could not reach a consistent consensus. For example, if two models predicted class A, two predicted class B, and one predicted class C, no majority decision would be possible.

Figure 3 illustrates the overall pipeline of the proposed framework. Specifically, given an ultrasound video clip, we first resize all frames to 256×256 and normalize them in [0, 1], resulting in $\mathbf{V} \in \mathbb{R}^{20\times 256\times 256}$. Then, each classifier independently predicts its corresponding finger motion class. The final prediction is made by aggregating these results.

Experimental setup and evaluation metrics

The study was conducted according to the guidelines of the Declaration of Helsinki and approved by the Institutional Review Board and Human Research Ethics Committee of Inje University (INJE 2022-12-001-001). All participants were provided informed consent during the study. Figure 4 shows the experimental setup for data acquisition. We used a hand-held ultrasound imaging system (Lumify L12-4, Philips Healthcare, Bothell, WA) with a linear array transducer. The transducer was positioned on the anterior compartment of the forearm.

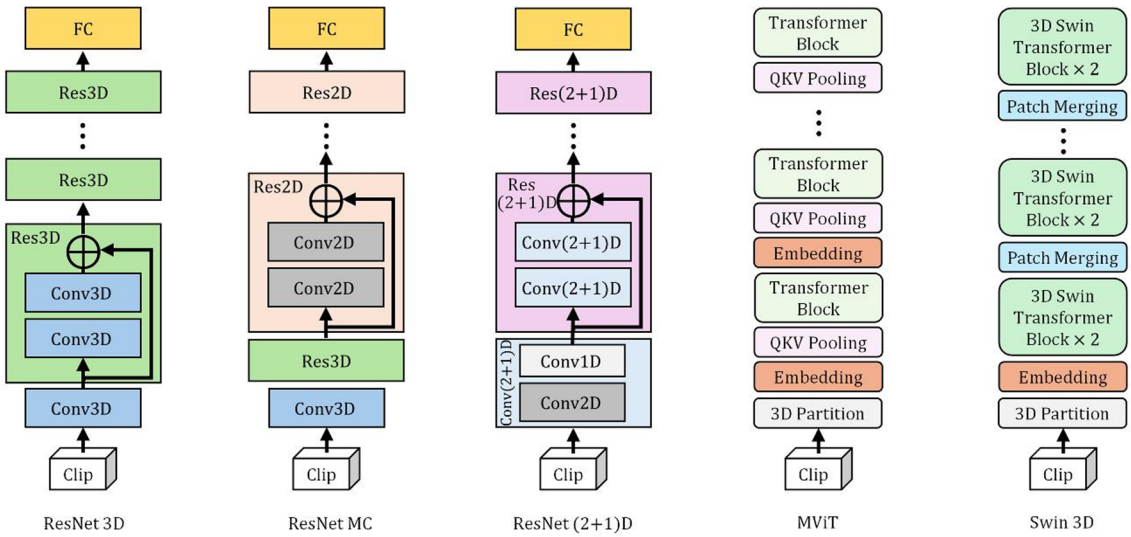


Fig. 2. The architectures of classifiers used for finger motion recognition.

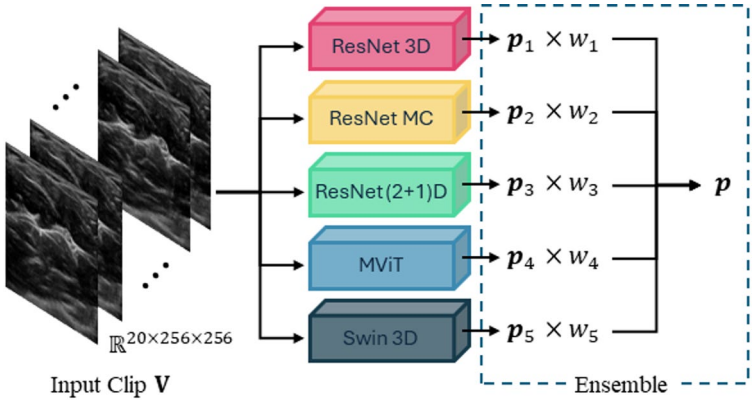


Fig. 3. Overall procedure of the proposed framework.



Fig. 4. Experimental setup for data acquisition. The participants were asked to move fingers following an instruction video.

A custom-made cuff was used to fix the transducer during the experiments, and acoustic gel was applied between the transducer and the skin to minimize acoustic impedance mismatching. To ensure the consistent position, the transducer was placed 60 mm away from the wrist, covering all muscles related to the finger movement.

The participants were asked to sit in front of a laptop and instructed to move their fingers, as shown in Fig. 4. We acquired ultrasound images from 30 able-bodied participants. The age range of participants was between 20 ~ 27 and sex distribution was 60:40 for male and female. All participants, except for two, were right-handed. They performed nine predefined finger motions (Fig. 5), including individual and multi-finger motions. Each motion was performed 20 times with both the left and right arms. During each trial, the motion was maintained for 3 s, followed by a 2-second rest interval, resulting in a total duration of 100 s per motion. Upon completion of 20 repetitions, participants were given a 1-minute rest period to avoid muscle fatigue. In addition, we acquired ultrasound images for both pronation and supination positions (see Fig. 4), resulting in a total of 2400 ultrasound video clips (30 × 80) for each motion. Unlike previous works that utilized A-mode ultrasound, the shift in transducer position did not significantly alter the accuracy of classification[14,16,17] since we utilized a whole 2D B-mode image. However, in some cases, it was observed that the transducer became detached from the skin during the experiments due to the deformation of forearm muscles. Thus, we excluded those images from network training and validation datasets. Table 1 summarized the number of ultrasound video clips used in this work. As summarized in Table 1, approximately 16% and 33% of the data were excluded. Transducer detachment occurred more frequently during pronation due to its hanging orientation. Training, validation, and test data were randomly divided across all video clips regardless of participants at a ratio of 6.5:1:2.5.

The frame rate of ultrasound imaging was 30 Hz, and each motion took around three seconds; thus, each video clip contained approximately 90 frames of ultrasound image for each motion. To reduce the computational

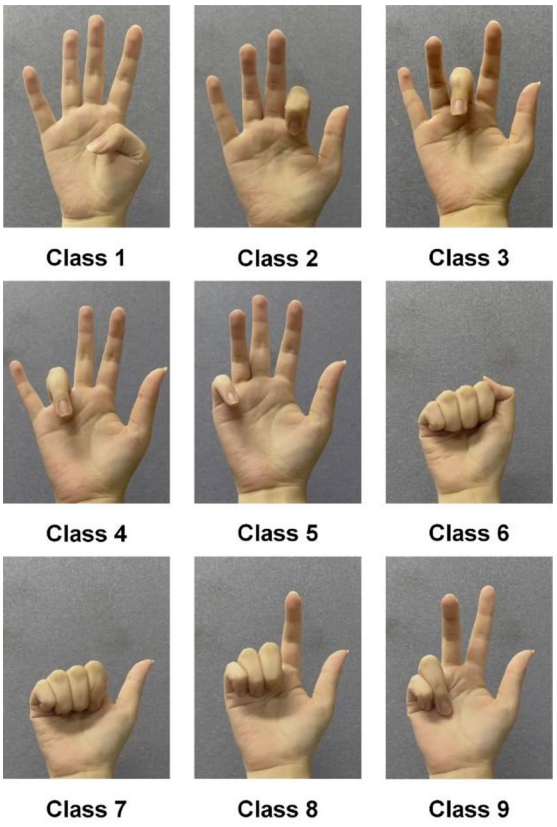


Fig. 5. Nine predefined finger motions including five single finger motions and four multi-finger motions.

Class	Supination	Pronation
1	949	779
2	996	803
3	1002	797
4	993	792
5	984	788
6	989	803
7	994	797
8	997	794
9	965	790

Table 1. Number of data used for training and validation.

burden, we uniformly split them into non-overlapped 20-frame images for all motions to train five classifiers. Although the participants followed an instruction video to perform the finger movements, the start and end points of each motion varied. Thus, we manually segmented the video clips to ensure consistent start points for training. Then, the images were resized to 256×256; thus, the input dimensions for training the network were $20 \times 256 \times 256$. No preprocessing techniques were applied, except for resizing and frame splitting, to preserve the raw characteristics of the ultrasound data.

For quantitative evaluation, classification accuracy (CA) and F1 score derived from a confusion matrix were computed[30]. CA is given by

$$CA = \frac{TP + TN}{TP + FN + FP + TN} \tag{2}$$

where TP, FN, FP, and TN stand for true positive, false negative, false positive, and true negative, respectively. F1 score is a weighted average of two competing metrics, i.e., precision and recall, leading to widespread used in machine learning evaluation. F1 score is computed as

$$F1\ score = 2 \times \frac{recall \times\ precision}{recall + precision} \tag{3}$$

$$Precision = \frac{TP}{TP + FP} \tag{4}$$

$$Recall = \frac{TP}{TP + FN} \tag{5}$$

Results

Real-time ultrasound imaging captured during the corresponding motions is provided in Supplementary video 1. As can be seen, the muscles undergo deformation as the finger motions are executed. Figure 6 presents the training and validation loss curves for each method. To prevent overfitting and to identify the optimal model, early stopping was applied based on the validation loss. Figure 7 shows a confusion matrix of the test set for classifying nine finger motions using the proposed method. The motion classes from 1 to 9 correspond to the predefined finger motions in Fig. 5. CA and F1 score derived from the confusion matrix were summarized in Table 2. The overall average CA and F1 scores were 95.64% and 0.9563, respectively. The average F1 score for the classification of individual finger motions (class 1 ~ 5) was 0.9745. For the classification of multi-finger motions, the average F1 score was slightly lower than for the individual cases (0.9336). The classification performance for class 7 was the lowest; CA and F1 scores were 92.37% and 0.9042, respectively. This can be attributed to the relatively small morphological changes in muscles related to the thumb (i.e., FPL) compared to other finger motions, leading to occasional misclassification and confusion with class 6. These experimental results suggest that the proposed framework effectively analyzes the forearm muscle activities and accurately classifies finger motions.

Table 3 presents the average performance metrics of the five classifiers, with ResNet 3D achieving the highest accuracy. The ensemble model outperformed the ResNet 3D classifier by approximately 0.01 in average performance metrics. Note that (1) MViT and Swin 3D have outperformed ResNet-based methods on various video recognition tasks by a large margin[26,27]; (2) ResNet 3D is inferior to ResNet MC and (2 + 1)D for action recognition on the Kinetics-400 dataset[31] from the previous study[24]. However, these tendencies observed in general video were completely opposite in ultrasound video clips. Specifically, the performance gaps between CNNs and transformers indicate that exploiting the fine-grained spatiotemporal correlation between adjacent pixels is essential to effectively analyze forearm muscle activities. While vision transformers are indeed capable of capturing long-range dependencies and global context, their reliance on non-overlapping patch partitioning may result in the loss of local spatial continuity[32,33]. This can be particularly limited in classifying finger motions in ultrasound clips, where subtle and local variations across frames encode discriminative patterns. Moreover, by comparing ResNet 3D and (2 + 1)D with ResNet MC, we see that temporal modeling is beneficial in the deep layers as well as the shallow layers.

In addition, we compared the computational efficiency of the models in terms of runtime, number of parameters, and the total number of floating-point operations (FLOPs) in Table 3. Here, the runtime was measured on a single NVIDIA RTX 4090 GPU. These results show that ResNet-based models are more efficient than transformer-based models even with a similar number of parameters.

Table 4 presents the results of a 4-fold cross-validation to assess the generalizability of the models[30]. Here, the dataset was evenly divided into four subsets; in each iteration, three subsets were used for training and the remaining one for validation. This process was repeated four times, ensuring that each subset was used once for

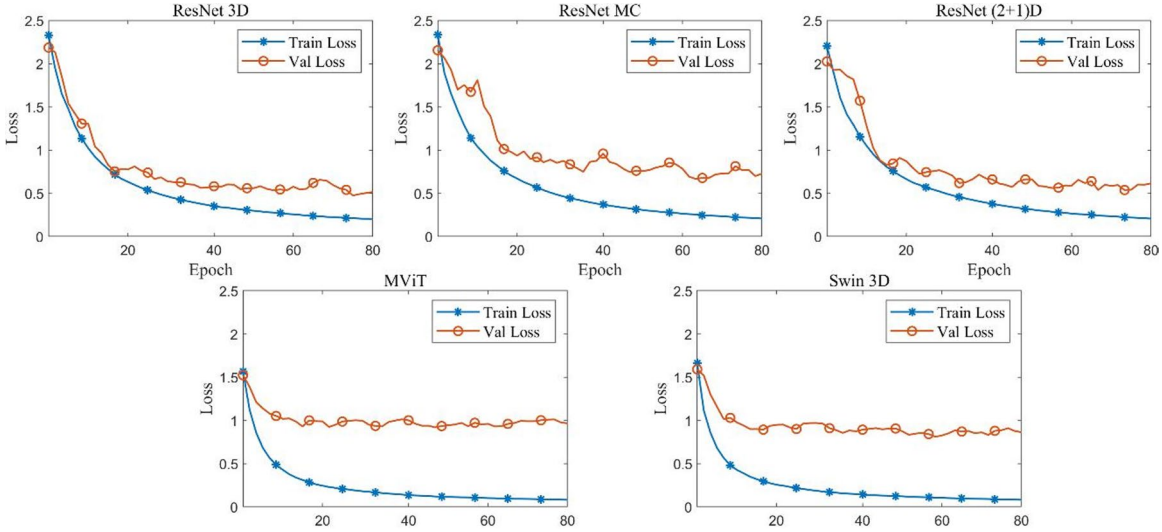


Fig. 6. Illustration of the training and validation loss curves.

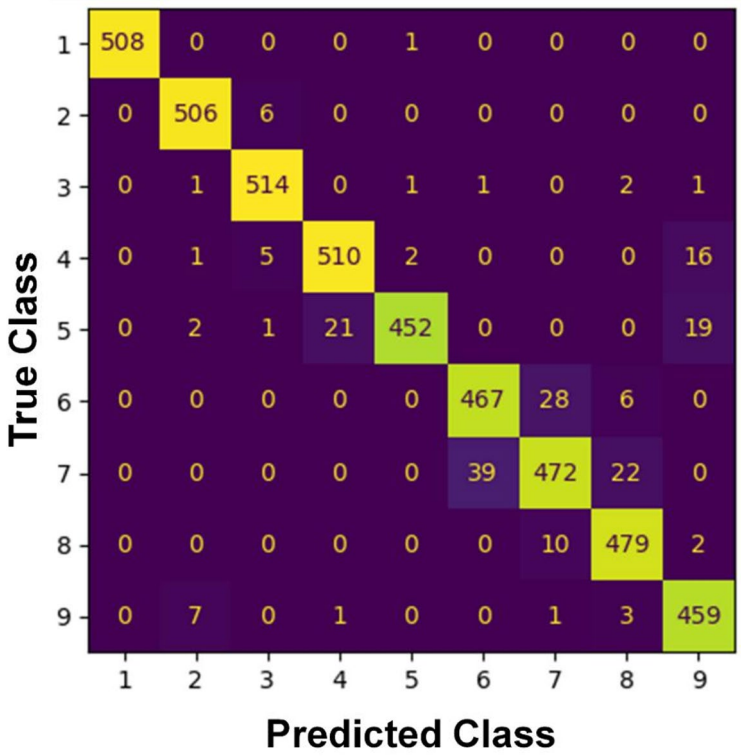


Fig. 7. Confusion matrix of test dataset for nine different finger motions based on the proposed method. Each motion corresponds to the predefined finger motions in Fig. 5.

Class		Accuracy	F1 score	Precision	Recall
Pronation	1	1.0000	0.9978	0.9956	1.0000
	2	1.0000	1.0000	1.0000	1.0000
	3	0.9960	0.9940	0.9921	0.9960
	4	0.9395	0.9569	0.9749	0.9395
	5	0.9906	0.9231	0.8642	0.9906
	6	0.9208	0.9465	0.9736	0.9208
	7	0.9791	0.9141	0.8571	0.9791
	8	0.9023	0.9487	1.0000	0.9023
	9	0.9004	0.9412	0.9858	0.9004
Supination	1	1.0000	1.0000	1.0000	1.0000
	2	0.9508	0.9618	0.9730	0.9508
	3	0.9526	0.9679	0.9838	0.9526
	4	0.9836	0.9556	0.9292	0.9836
	5	0.9920	0.9845	0.9771	0.9920
	6	0.9215	0.8991	0.8778	0.9215
	7	0.8512	0.8856	0.9229	0.8512
	8	0.9790	0.9610	0.9437	0.9790
	9	0.9537	0.9567	0.9597	0.9537

Table 2. Performance metrics for nine different finger motions under pronation and supination positions using the proposed framework.

validation. The results demonstrated that the models achieved robust generalization. Moreover, the ensemble model consistently outperformed individual models across all folds, highlighting the reliability of the proposed approach.

Table 5 summarizes the performance of conventional machine learning approaches. In this evaluation, we implemented three widely used classifiers[30]: Decision Tree, Support Vector Machine (SVM), and LDA. For feature extraction, two types of handcrafted descriptors were used: a PCA-based extractor[17] and a Local Binary

Classifier	Accuracy (%)	F1 score	Parameters	GFLOPS	Runtime (ms)
ResNet 3D	94.68	0.9466	33.4 M	40.70	1.3
ResNet MC	92.66	0.9268	11.7 M	43.34	1.2
ResNet (2 + 1)D	94.26	0.9423	31.5 M	40.52	2.2
MViT	86.07	0.8609	34.5 M	64.22	495
Swin 3D	84.80	0.8487	28.2 M	43.88	8.4
Ensemble	95.66	0.9563	-	-	-

Table 3. Comparison of classifier performance.

Accuracy (%)	1 st fold	2nd fold	3rd fold	4th fold
ResNet 3D	91.39	94.71	93.53	93.58
ResNet MC	92.86	93.56	88.14	91.12
ResNet (2 + 1)D	94.26	94.13	94.45	89.68
MViT	87.12	88.19	86.22	85.26
Swin 3D	88.34	90.03	84.89	84.10
Ensemble	95.62	95.68	95.29	95.05

Table 4. Results of 4-fold cross-validation.

Feature extractor	Classifier	Accuracy (%)	F1 score	Runetime (ms)
PCA-based extractor	Decision Tree	31.58	0.3141	164
	SVM	32.89	0.3276	164
	LDA	31.00	0.3098	165
LBP-based extractor	Decision Tree	27.31	0.3720	100
	SVM	22.29	0.2197	101
	LDA	37.71	0.3743	102
Proposed ensemble		**95.66**	**0.9563**	

Table 5. Performance comparison of the proposed method with conventional machine learning methods.

Pattern (LBP)-based descriptor[34]. The results indicate that these conventional methods yielded significantly lower performance compared to the proposed deep learning approach, highlighting the limitations of traditional classifiers and the advantages of deep models.

We also conducted analyses to better understand the impact of model design choices and training strategies. First, we compared 2D and 3D CNN architectures to assess the effect of temporal modeling. In this experiment, we employed 2D ResNet-18 with stacked ultrasound images along the channel dimension. We observed that 2D ResNet-18 achieved lower performance (accuracy: 81.85%, F1 score: 0.8157) than 3D ResNet-18 (accuracy: 94.68%, F1 score: 0.9466). This experiment highlights the importance of explicitly capturing short-term motion dynamics in ultrasound video for improved classification performance.

In addition, we analyzed the impact of different ensemble voting schemes on classification performance. Using unweighted voting, the ensemble achieved an accuracy of 94.66%. In contrast, the weighted voting strategy described in Eq. 1 improved performance to 95.66% by effectively resolving prediction conflicts among individual models. To further assess the robustness of the proposed method, each model was trained 10 times with different random seeds. The resulting ensemble models achieved accuracy ranging from 95.43% to 96.19%, with an average accuracy of 95.92% and a deviation of 00.24%, demonstrating consistent and reliable performance.

Discussion

In this work, we implemented and evaluated five representative deep video classification architectures: ResNet 3D, ResNet MC, ResNet (2 + 1)D, MViT, and Swin 3D. These models widely recognized as standard baselines or backbone networks video-based analysis. Although various state-of-the-art models have been proposed, most of them are built upon derivative structures of these foundational models. Therefore, using these core backbones provides a more objective basis for performance evaluation. Based on these results, we proposed a deep learning-based framework for classifying finger motions in ultrasound-based HMI. The experimental results demonstrated that the proposed approach achieves high classification accuracy. The effectiveness of individual neural network modules for finger motion recognition was validated using ultrasound video sequences. In addition, the feasibility of the method was confirmed by acquiring ultrasound data in both pronation and supination postures, showing its potential for robust and versatile application in ultrasound-based HMIs.

A previous study demonstrated that 1D A-mode ultrasound outperformed conventional sEMG-based methods in terms of finger motion recognition accuracy[17]. In this work, they placed the four transducers to cover FDS, FDP, FPL, EDC, and EPL, which are crucial for finger movement. While A-mode ultrasound provides advantages such as affordability, size, and computational efficiency, its performance heavily depends on the precise placement of the transducer. In addition, wrist rotations (pronation and supination), which induce dynamic shifts in both flexor and extensor muscle positioning within the forearm, and the slight shift of the transducer can significantly degrade the accuracy of A-mode ultrasound-based HMI, posing a significant challenge to its robustness and reliability[13,17]. Unlike A-mode, the comprehensive 2D visualization of muscle activity in B-mode ultrasound eliminates the dependence on precise transducer placement, rendering its performance stable and consistent across wrist rotations. In addition, recent breakthroughs in wearable ultrasound technology could facilitate B-mode ultrasound-based HMI by overcoming its limitations in portability and ease of use[35,36].

Our study has some limitations. First, while we collected an extensive dataset by recording 20 repetitions of each motion from both arms, including both pronation and supination positions, the overall number of participants remains insufficient for achieving robust classification performance. Increasing the amount of data can significantly enhance the performance of deep learning models, resulting in higher accuracy and improved robustness. In addition, rather than randomly dividing all video clips regardless of participant, future studies will adopt participant-wise partitioning to mitigate the risk of data leakage and ensure more reliable model evaluation for diverse applications. The diversity of the dataset is also critical for achieving better generalization in deep learning models. This ensures that the model can handle unseen datasets more effectively and prevents overfitting to specific patterns.

Another limitation is that our feasibility study only classified nine predefined finger movements (five single-finger motions and four multi-finger motions). To develop a more versatile solution, it is essential to include more complex combinations of finger movements in the model training. This will enhance the robustness of the proposed framework. Alternatively, the appropriate architecture modification from the conventional one may be necessary to accommodate the increased complexity and variability in finger movements. Training deep learning classification models on a more diverse finger movement dataset enhances generalization and overall performance, as these methods inherently learn features without manual preprocessing such as filtering or PCA. We are planning to collect more datasets and classify a broader range of dexterous hand gestures. In addition, we focused only on the forearm muscles that are primarily involved in large finger movements. Fine control of single finger movement and grip force is produced by the intrinsic muscles located within the hand[37]. Thus, to accurately recognize dexterous hand motions, it may be necessary to monitor both the forearm muscles and the intrinsic muscles, which will be investigated in the following study. Lastly, we did not implement the proposed method in real time. Real-time capability is crucial for the control of hand prostheses. Despite the relatively shallow architecture of the proposed method, the computational complexity of 3D or (2 + 1)D convolution remains a challenge for achieving real-time operation on mobile/embedded devices. A previous study recommended an optimal control delay of 100 ~ 125ms for myoelectric prostheses[38]. Recently, mobile deep learning frameworks such as TensorFlow lite and Caffe2 have been developed, opening up a new era of on-device intelligence[39]. We believe that the proposed method could be easily implemented on low power mobile/embedded devices, which will be investigated in the future as well.

In the future work, we plan to conduct direct comparisons of finger motion recognition performance among the proposed method, 1D A-mode ultrasound, and sEMG-based approaches under identical conditions. Such analyses would clarify the strengths and weaknesses of each method and may provide new insights for advancing gesture recognition for many applications. In addition, although ultrasound images in this study were acquired under controlled conditions (e.g., fixed posture and fixed transducer position), we aim to extend the training the proposed method to more variable conditions, including free posture, varying transducer positions, and more complex combinations of finger movements. For this, we are currently developing a wearable ultrasound imaging system. Moreover, we plan to recruit amputee participants for future experiments, which will further enhance the applicability of the proposed method in practical scenarios. In addition, each motion lasted approximately 3 s, a duration intentionally selected to ensure consistency across participants and minimize segmentation errors. As this study focused on large finger motions, 20 frames per clip were considered sufficient for reliable classification. Although shorter or overlapping clips may improve recognition of subtle motions, the current design provides a stable basis for feasibility evaluation, and future work will extend to subtle motions. Finally, muscle fatigue, tremors, and involuntary movements may affect performance. In this study, such cases were excluded to provide a reliable foundation for evaluating the feasibility of ultrasound-based motion recognition. Future work will incorporate these scenarios to enhance real-world applicability.

Conclusion

In this study, we proposed a framework using ultrasound imaging for finger motion classification. The experimental results demonstrated the feasibility of the proposed method in accurately classifying various finer movements, including individual and multi-finger motions. This approach would offer notable advantages compared to conventional sEMG-based HMI. We believe that the proposed method could be used for diverse applications such as gesture recognition in VR/AR, controlling robotic hands, and assisting in medical rehabilitation for finger dexterity.

Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

Appendix

Conventional ML methods

We implement several conventional machine learning classifiers. First, features are extracted using two different methods: a PCA-based extractor[17] and a Local Binary Pattern (LBP)-based descriptor[34]. These features are then used to train classic classifiers[27] of Decision Tree, Support Vector Machine (SVM), and Linear Discriminant Analysis (LDA).

For the PCA-based feature extractor[17], we select five columns from each ultrasound frame and divide each column into multiple non-overlapping vertical windows of 20 pixels. For each window, we calculate the slope of the best-fit line using linear regression. The slope coefficients from all windows and all five columns are then concatenated into a single feature vector, which is subsequently reduced to 12 dimensions using PCA.

The LBP-based descriptor[33] is implemented to obtain local texture features in ultrasound images. For each frame, an LBP map is computed by comparing each center pixel with its neighboring pixels, and then it is converted into a normalized histogram with 32 bins. Finally, the histograms from all 20 frames are concatenated to form a feature vector for each ultrasound clip.

All conventional classifiers are implemented using the Scikit-Learn tool[40]. They are trained and evaluated using the same training and test splits as used in the proposed framework for fair comparison.

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Yohan Lee: Conceptualization, Validation. Keunsoo Ko: Software, Investigation, Writing – review & editing. Jaeyeop Jang: Data acquisition, Software. Jung-Min Lee: Interpretation of data, Validation. Changhan Yoon: Writing – original draft, Validation, Writing – review & editing, Supervision, Funding acquisition.

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Declarations

Competing interests

The authors declare no competing interests.

Ethics declarations

The study was conducted according to the guidelines of the Declaration of Helsinki and approved by the Institutional Review Board and Human Research Ethics Committee of Inje University (INJE 2022-12-001-001). All participants were provided informed consent during the study.

Additional information

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